

Kittichai Panyapa

กิตติชัย ปัญญาปา

Unity3D Gameplay & Multiplayer Programmer

Unity3D / C# | Multiplayer Networking (NGO → transferable to Photon) | Relay & Matchmaking | DOTween & UI



ABOUT ME

Game design graduate (Interactive Design & Game, Faculty of Digital Media, Sripatum University) specializing in Unity3D and C#, with hands-on experience shipping a 3D co-op asymmetric multiplayer game. Implemented a server-authoritative multiplayer architecture (NetworkVariable/NetworkList, ServerRpc/ClientRpc, Relay server with join-code matchmaking) and designed switch/mechanic systems kept in sync across clients in real time. Comfortable with OOP, interface abstraction, event-driven design, clean code, and performance optimization (MaterialPropertyBlock).

Networking background is built on Unity Netcode for GameObjects (NGO) rather than Photon directly — but the underlying patterns transfer directly: server authority, RPC-style calls (ServerRpc/ClientRpc ↔ Photon RPCs), and synchronized network state. Ready to ramp up on Photon's API (PUN2 / Fusion / Quantum) quickly given this foundation.

TECHNICAL SKILLS

Engine & Tools	Unity3D, C# Scripting, DOTween, Unity New Input System, Unity Localization, Cinemachine
Multiplayer Networking	Multiplayer Architecture (Host/Client), Server-Authoritative Design, NetworkVariable/NetworkList, ServerRpc/ClientRpc, Relay Transport & Join-Code Matchmaking, Networked Scene Loading — patterns transferable to Photon (PUN/Fusion)
Architecture & Gameplay	Object-Oriented Programming (OOP), Component-based Design, Interface Abstraction (5+ custom interfaces), State Management, Event-Driven Architecture
Optimization & Debugging	Draw Call Optimization (MaterialPropertyBlock), Frame Rate Management, Physics Optimization, In-Game Developer Console
Team & Workflow	Git, GitHub/GitLab Workflow, Code Review, Cross-discipline Collaboration with Designers & Artists

FEATURED GAME DEVELOPMENT PROJECT

[Together in the Void ↗](#) (3D Platformer Co-op, 2 Players) — Senior Thesis Project

Lead Gameplay & Network Programmer | 2025-2026

Tech Stack: Unity3D, C#, Unity Netcode for GameObjects (NGO), Unity Relay / UTP, DOTween, Unity Localization

- Multiplayer Networking & Matchmaking:** Designed a real-time host/client system over Unity Relay with join-code matchmaking, automatic room entry, character-slot assignment, and synchronized scene loading between host and client.
- Networked Interaction & Puzzle Mechanics:** Built co-op mechanics (interactive switches, multi-zone wind system) behind an ISwitchableWindManager interface, syncing state through server-authoritative NetworkVariables to prevent desync between players.
- 3D Physics & Movement:** Implemented Rigidbody-based real-time movement and physics interactions — wind zones, moving platforms, checkpoints, and traps.
- Modular World UI & Localization:** Built world-space UI interactions with DOTween-animated feedback, Sphere Overlap detection, player-freeze/cursor-lock handling, and TH/EN localization.
- Optimization & Developer Tooling:** Reduced draw calls with MaterialPropertyBlock instead of per-object material instances; built an in-game developer console (~ key) for fast host/join testing.
- Team Collaboration:** Sole programmer on a 6-person team (1 game designer, 2 2D artists, 1 3D artist, 1 animator/rigger) — owned 100% of the gameplay and networking code, working most closely with the game designer to translate mechanics into working systems.

EDUCATION

Bachelor's Degree, Interactive Design & Game

Faculty of Digital Media • Sripatum University | GPA: 3.62 / 4.00

CONTACT

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